(Please write your Enrollment Number) Enrollment No. 194

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech>, Semester:<5>
(December, 2025)

Subject Code: BCS 303	Subject: Theory of Computation
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)
	a) What is ε-closure of a state? What are the differences among DFA, NFA and ε-NFA	
	b) What is regular grammar and what are the properties of regular languages	
	c) Give the Formal definition of CFG and what is parse tree	
	d) What is useful and useless symbols in a grammar? Give examples	
	e) What is Linear Bounded Automata? How it is different from Finite Automata.	
	f) Give the formal definition of Turing Machine	
	g) Give the formal definition of Mealy Machine. How Mealy machine is different from Moore machine	
	h) What is halting problem	
	UNIT I	
Q2	Convert the following NFA to optimized DFA	(10)
Q3	Show that the following language is not regular using pumping lemma $L = \{a^{k^2} / k \geq 1\}$	(10)
	UNIT II	
Q4	Find a Greibach normal form grammar equivalent to the following CFG: S→AA| 0 A→ SS| 1	(10)
Q5	Design a PDA for the language $L = \{a^{2m} b^{3n} c^{3n} d^{2m} \mid m, n \geq 1\}$	(10)

δ	a	b	c	d
q_0	--	--	--	$\{q_0, q_1\}$
q_1	--	--	$\{q_1, q_2\}$	--
q_2	--	$\{q_2, q_3\}$	--	--
q_3	$\{q_3, q_0\}$	--	--	--

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	UNIT III	
Q6	Design a Turing Machine for the language: $L = \{a^n b^n c^{2n} / n \geq 1\}$	(10)
Q7	Show that the following language is not CFL $L = \{a^{2^k} / k \geq 1\}$	(10)
	UNIT IV	
Q8	(a) Explain about P and NP. NP complete & NP Hard Problems with examples. (b) Explain Post correspondence Problem with suitable example.	(5,5)
Q9	Design a Moore machine for the following process: for input from (0+1)*, if the input ends in 101, out A; if the input ends in 110, output B; otherwise output C.	(10)

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